

DAA LAB ASSIGNMENT 2

Name – Nandini Yadav

Section-c

Semester -5th (3rd year)

Que-1 write a program for insertion sort

```
#include<stdio.h>
```

```
int main(){
```

```
int n,a[10],i,j,key;
```

```
printf("enter the size of array");
```

```
scanf("%d",&n);
```

```
printf("enter the elements of array ");
```

```
for(i=0;i<n;i++){
```

```
    scanf("%d",&a[i]);
```

```
}
```

```
for(j=1;j<n;j++){
```

```
    key=a[j];
```

```
    i=j-1;
```

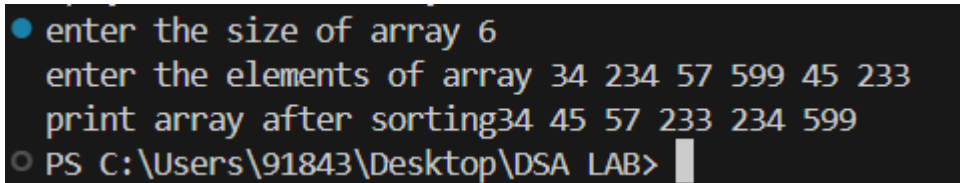
```
    while(i>=0 && a[i]>key){
```

```
        a[i+1]=a[i];
```

```

        i=i-1;
    }
    a[i+1]=key;
}
printf("print array after sorting");
for(i=0;i<n;i++){
    printf("%d ",a[i]);
}
return 0;
}

```



```

● enter the size of array 6
  enter the elements of array 34 234 57 599 45 233
  print array after sorting 34 45 57 233 234 599
○ PS C:\Users\91843\Desktop\DSA LAB>

```

Que -2 write a program for selection sort

```

#include <stdio.h>

void selection(int arr[], int n)
{
    int i, j, small;

    for (i = 0; i < n-1; i++)

```

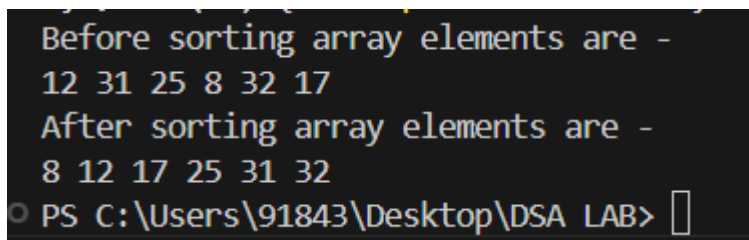
```
{  
    small = i;  
  
    for (j = i+1; j < n; j++)  
        if (arr[j] < arr[small])  
            small = j;  
  
    int temp = arr[small];  
    arr[small] = arr[i];  
    arr[i] = temp;  
}  
}
```

```
void printArr(int a[], int n)  
{  
    int i;  
    for (i = 0; i < n; i++)  
        printf("%d ", a[i]);  
}
```

```

int main()
{
    int a[] = { 12, 31, 25, 8, 32, 17 };
    int n = sizeof(a) / sizeof(a[0]);
    printf("Before sorting array elements are - \n");
    printArr(a, n);
    selection(a, n);
    printf("\nAfter sorting array elements are - \n");
    printArr(a, n);
    return 0;
}

```



```

Before sorting array elements are -
12 31 25 8 32 17
After sorting array elements are -
8 12 17 25 31 32
PS C:\Users\91843\Desktop\DSA LAB>

```

Que -3 Write a program to implement quick sort

```
#include<stdio.h>
```

```
void swap(int* a, int* b) {
```

```
    int t = *a;
```

```

    *a = *b;

    *b = t;
}

int partition(int arr[], int low, int high) {
    int pivot = arr[high];
    int i = (low - 1);

    for (int j = low; j <= high - 1; j++) {
        if (arr[j] < pivot) {
            i++;
            swap(&arr[i], &arr[j]);
        }
    }

    swap(&arr[i + 1], &arr[high]);
    return (i + 1);
}

void quickSort(int arr[], int low, int high) {
    if (low < high) {
        int pi = partition(arr, low, high);
        quickSort(arr, low, pi - 1);
    }
}

```

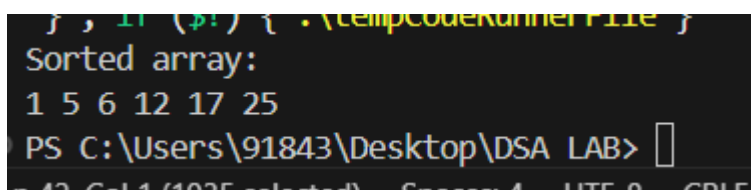
```

        quickSort(arr, pi + 1, high);
    }
}

void printArray(int arr[], int size) {
    int i;
    for (i = 0; i < size; i++)
        printf("%d ", arr[i]);
    printf("\n");
}

int main() {
    int arr[] = { 12, 17, 6, 25, 1, 5 };
    int n = sizeof(arr) / sizeof(arr[0]);
    quickSort(arr, 0, n - 1);
    printf("Sorted array: \n");
    printArray(arr, n);
    return 0;
}

```



```

    } , if ($?) { .\tempCodeRunnerFile }
Sorted array:
1 5 6 12 17 25
PS C:\Users\91843\Desktop\DSA LAB>

```

